

Diabetes Risk Prediction

Project Report

AI/ML Internship Program — Capstone Project (Weeks 5–6)

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1. Introduction

Diabetes is one of the most common chronic conditions worldwide, and early detection meaningfully improves patient outcomes by allowing intervention before complications develop. In many primary-care and low-resource settings, however, full diagnostic testing is not always immediately available, which creates a real gap between when risk factors are present and when they are actually caught.

This project addresses that gap with a machine learning screening tool: a trained classifier that estimates a patient's diabetes risk from eight routine diagnostic measurements, wrapped in a simple web application so the prediction is usable by a non-technical person rather than locked inside a notebook. The project covers the full applied ML workflow — exploratory data analysis, data cleaning, feature engineering, model training and comparison, threshold tuning, robustness testing, and deployment as a working application.

2. Problem Statement

Build a supervised machine learning model that predicts whether a patient is likely to have diabetes based on routine diagnostic health measurements, and surface the key risk factors driving each prediction. The model should be usable as an early, low-cost screening signal — not a diagnosis — in contexts where full diagnostic testing isn't always immediately available.

2.1 Dataset

The Pima Indians Diabetes Dataset contains 768 patient records with 8 diagnostic features and a binary target (Outcome: 1 = diabetic, 0 = non-diabetic).

Feature	Description
Pregnancies	Number of times pregnant
Glucose	Plasma glucose concentration (2-hr oral glucose tolerance test)
BloodPressure	Diastolic blood pressure (mm Hg)
SkinThickness	Triceps skin fold thickness (mm)
Insulin	2-hour serum insulin (mu U/ml)
BMI	Body mass index
DiabetesPedigreeFunction	Diabetes likelihood based on family history
Age	Age in years
Outcome (target)	1 = diabetic, 0 = non-diabetic

2.2 Success Metrics

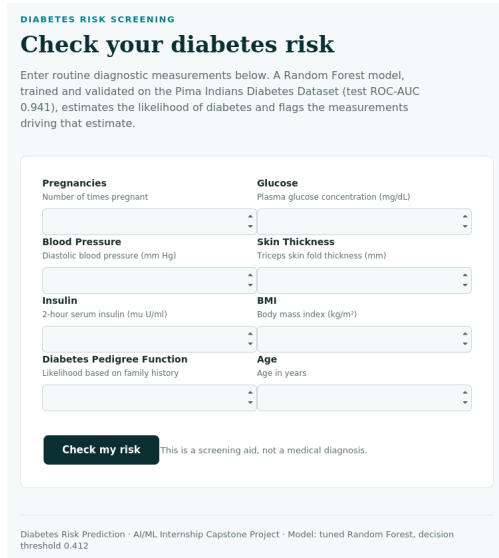
The primary evaluation metrics are recall on the diabetic class and ROC-AUC, prioritized over raw accuracy. A false negative — telling a diabetic patient they are low-risk — is the costlier error in a screening context, so accuracy alone would be a misleading measure of model quality on this ~65/35 imbalanced dataset. Target: ROC-AUC > 0.80.

3. Application Overview & UI Screenshots

The trained model is served through a small Flask web application, so a prediction is a form submission away rather than a notebook cell run. The app takes the same 8 raw measurements used in training, re-derives the engineered features from Sprint 1 internally, and returns a risk classification, a probability, and the top contributing factors for that specific patient.

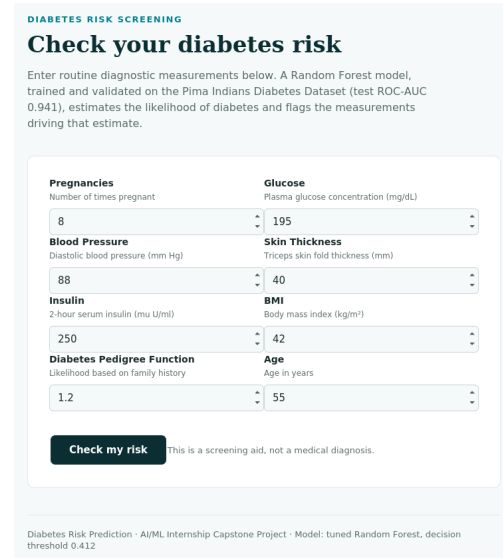
3.1 Input Form

The landing page collects the 8 diagnostic measurements, each labeled in plain language with its clinical meaning.



The screenshot shows the 'DIABETES RISK SCREENING' form titled 'Check your diabetes risk'. It includes a brief explanation of the model and a form with eight input fields arranged in two columns. The fields are: Pregnancies (Number of times pregnant), Glucose (Plasma glucose concentration (mg/dL)), Blood Pressure (Diastolic blood pressure (mm Hg)), Skin Thickness (Triceps skin fold thickness (mm)), Insulin (2-hour serum insulin (mu U/ml)), BMI (Body mass index (kg/m²)), Diabetes Pedigree Function (Likelihood based on family history), and Age (Age in years). Each field has a numeric input with up and down arrows. At the bottom, there is a 'Check my risk' button and a disclaimer: 'This is a screening aid, not a medical diagnosis.' The footer text reads: 'Diabetes Risk Prediction · AI/ML Internship Capstone Project · Model: tuned Random Forest, decision threshold 0.412'.

Figure 1. The diabetes risk screening form.



This screenshot shows the same form as Figure 1, but with sample patient data entered. The values are: Pregnancies: 8, Glucose: 195, Blood Pressure: 88, Skin Thickness: 40, Insulin: 250, BMI: 42, Diabetes Pedigree Function: 1.2, and Age: 55. The 'Check my risk' button and disclaimer are still present at the bottom. The footer text is identical to Figure 1: 'Diabetes Risk Prediction · AI/ML Internship Capstone Project · Model: tuned Random Forest, decision threshold 0.412'.

Figure 2. The form populated with a sample patient's measurements.

3.2 Prediction Results

Submitting the form returns a result page showing the predicted risk category, the model's estimated probability plotted against the tuned decision threshold, the top contributing factors for that patient, and a restatement of the measurements entered — so the person can verify what was submitted. Note the screening disclaimer's wording itself changes depending on the result, rather than showing a generic message regardless of outcome.

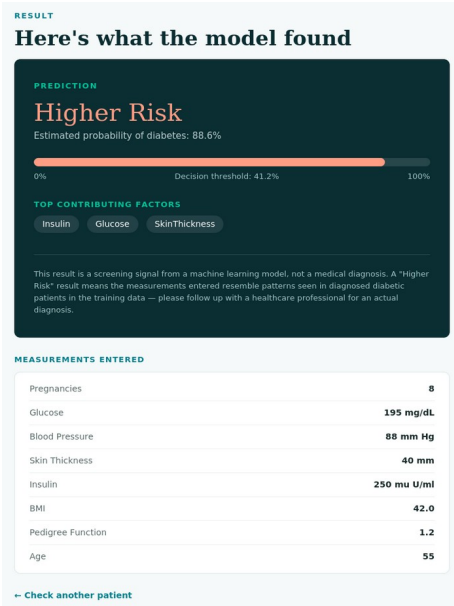


Figure 3. "Higher Risk" — elevated Glucose/BMI/Insulin drive an 88.6% estimated probability, above the 41.2% threshold.

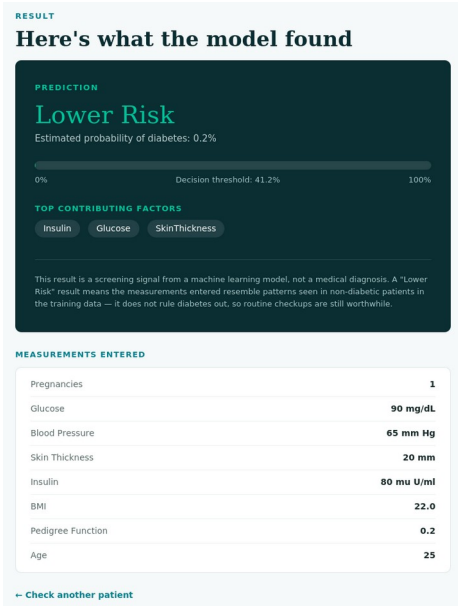


Figure 4. "Lower Risk" — a younger patient with measurements closer to the non-diabetic population (0.2%).

4. Results

4.1 Data Cleaning & Feature Engineering

Glucose, BloodPressure, SkinThickness, Insulin, and BMI contained biologically-impossible zero values (used as an implicit "not recorded" marker). These were converted to explicit missing values and imputed using the median within each Outcome group, preserving the real distributional difference between diabetic and non-diabetic patients. Four engineered features were added: BMI_Category and Age_Group (clinical bins), and Glucose_BMI_Interaction and Insulin_Glucose_Ratio (clinically-motivated interaction/ratio terms), both of which ranked among the strongest correlates with the target.

4.2 Model Comparison

Four classical models were trained and evaluated, with Logistic Regression and Random Forest tuned via GridSearchCV (5-fold stratified cross-validation); a Dropout/EarlyStopping-regularized Keras neural network was trained as a deep learning comparison point.

Model	Recall	Precision	ROC-AUC
Baseline (majority class)	0.000	—	0.500
Logistic Regression (tuned)	0.648	0.636	0.835
Keras Neural Network	0.852	0.648	0.880
Random Forest (tuned, selected)	0.796	0.811	0.941

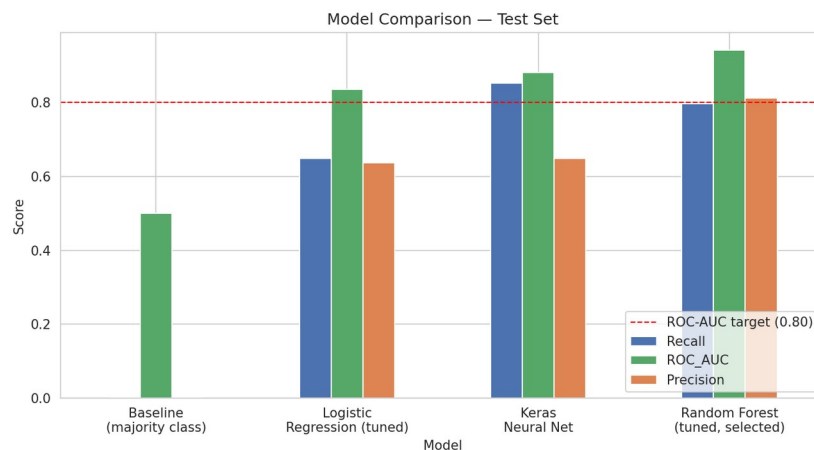


Figure 5. Test-set Recall, ROC-AUC, and Precision across all four models.

The Keras network achieved marginally higher recall than the tuned Random Forest, but at a meaningfully lower ROC-AUC and precision — a 6-point ROC-AUC gap reflects a real difference in overall discriminative power, and the network would flag substantially more healthy patients as at-risk. The tuned Random Forest was selected as the final model.

4.3 Threshold Tuning

Lowering the classification threshold from the default 0.5 to 0.412 — the lowest value achieving at least 85% recall — allows the Random Forest to match the Keras network's recall (0.852) while retaining substantially better precision (0.793 vs. 0.648).

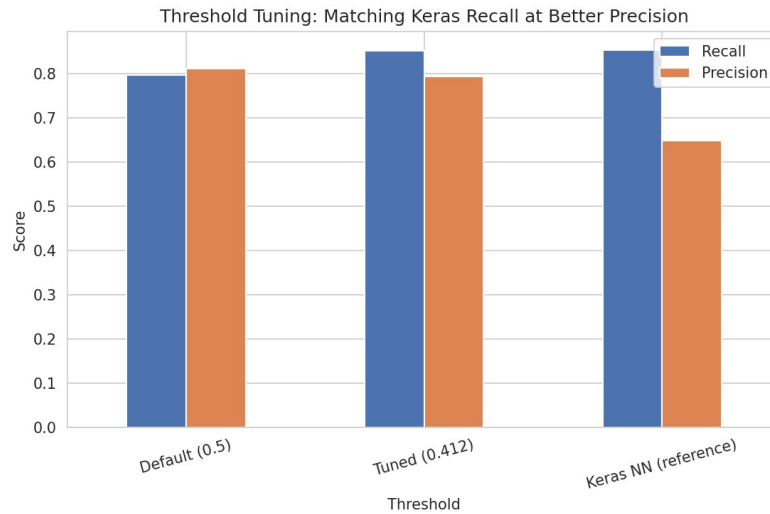


Figure 6. Threshold tuning closes the recall gap with the neural network without sacrificing precision.

4.4 Final Model Performance

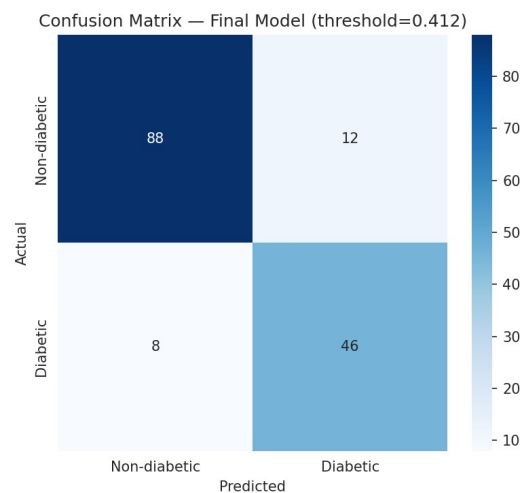


Figure 7. Confusion matrix at the tuned decision threshold (0.412) on the held-out test set ($n=154$).

A 500-iteration bootstrap resample of the test set produced a 95% confidence interval of [0.901, 0.972] around the 0.941 test ROC-AUC, indicating the result is stable rather than an artifact of a single train/test split.

5. Future Approach

The current model and application meet the goals set out in the problem statement, but several directions would meaningfully strengthen the project beyond this capstone:

- Validate on external data: test the model against a second, independently-collected diabetes dataset to check whether performance holds outside the Pima population the model was trained on.
- Investigate Insulin's feature importance: Insulin ranked as the top feature in the trained model despite requiring imputation for 48.7% of its values — this should be validated against data with genuinely observed insulin readings before being treated as a reliable clinical signal.
- Expand the subgroup analysis: the age-band performance check in this project was limited by a 154-row test set; a larger sample would allow a genuine fairness assessment rather than a directional one.
- Add a plain-language explanation layer: use an LLM to translate a prediction and its contributing factors into a short, plain-language explanation for a non-technical end user, building on the API integration skills covered in Week 4.
- Deploy publicly: move the Flask application from local execution to a hosted platform (e.g. Render, Railway, or a small cloud VM) so it is reachable via a live link rather than only runnable locally.
- Add input validation and clinical guardrails: real deployment would need stronger input validation, rate limiting, and a clearer escalation path directing "Higher Risk" users toward actual care rather than just a disclaimer.

6. Conclusion

The final Random Forest model, evaluated at a tuned decision threshold of 0.412, achieves 0.852 recall, 0.793 precision, and 0.941 ROC-AUC on the held-out test set — comfortably exceeding the target of ROC-AUC > 0.80 set at the start of the project. Wrapped in a working Flask application, the model moves from a notebook result to something an actual user could open and use, which was the point of building it as an application rather than stopping at a trained model.